CS 791 - Assignment 1 Report

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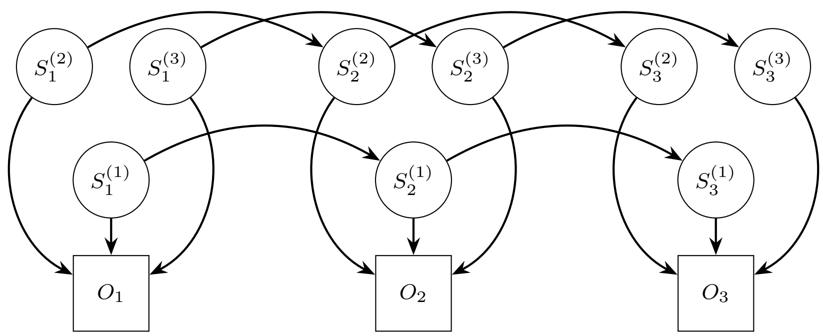
1 Factorial Hidden Markov Model (FHMM)

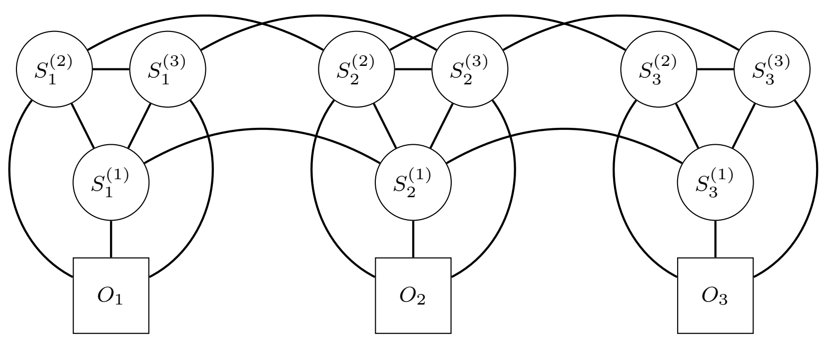
An FHMM is an extension of the standard Hidden Markov Model (HMM) that allows for multiple independent hidden state sequences to influence the observed data.

The problem of interest is to infer the marginals of the hidden states, and the most probable sequence of hidden states, given a sequence of observations in a Factorial Hidden Markov Model (FHMM).

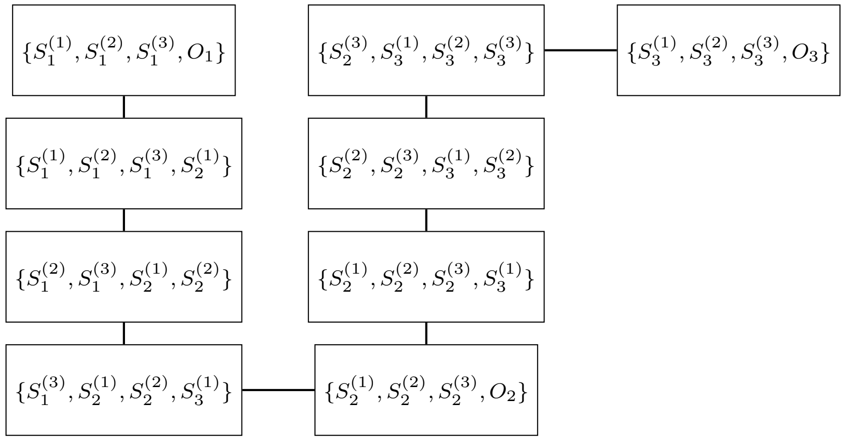
The problem involved making a junction tree out of the FHMM by triangulating it and then using the message passing algorithm to infer the marginals of the hidden states and the most probable sequence of hidden states given a sequence of observations.



The moralized graph looks like



The junction tree made out of the cliques after triangulation is a chain as shown below:



The message passing algorithm is now simple with just two passes, one from left to right and the other from right to left.

For including the prior of the observations, we merged the observations into the cliques by removing the observation nodes and considering the potentials over the cliques that include the observations.

In the junction tree, there are $M(T-1)+T$ cliques of size $M+1$ and all the cliques form a chain. Here, M is the number of factors and T is the number of time steps.

1.1 Z-Value Computation

For finding the z-value of the junction tree, the message passing algorithm runs over every clique for every assignment of the variable in the clique at max and hence the time complexity is $O(TMK^{M+1})$, assuming the translation of dict to the indices occurs in $O(1)$, where K is the number of states each hidden variable can take.

1.2 Marginals of the Hidden States

For the marginals of the hidden states, we ran the message passing algorithm forward and backward, and then computed the marginals for each hidden state by marginalizing over the cliques that contain the hidden state. This took $O(TMK^{M+1})$ time.

1.3 Most Probable Sequences of Hidden States

For the top-k most probable sequence of hidden states, we modified the message passing algorithm to each potential storing a k-heap of assignments and their probabilities and merging the k-heaps appropriately during message passing. This took $O(TMK^{M+1}k\log k)$ time.